ABT RECERTIFICATION EXAMINATION #23

May 5, 2006 *PART A*

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1997 and 2002. There are three parts to the exam. An answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading by **June 30, 2006.** If you wish to return them by courier, the physical address is 9200 Sanctuary Court, Raleigh, NC 27617. If you do not achieve a passing score on any of the parts, you will be contacted and given another opportunity to pass. If you pass all three parts, your scores will be provided to you in November of this year, along with the results of the Recertification Committee's review of your other application materials. If you have any questions please contact the ABT office at (919) 841-5022 or info@abtox.org.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.

2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.

3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

Key for the following questions
Choose the single most appropriate answer

1. Which of the following classes of pesticides exhibit Type I (T syndrome - aggressive behavior, whole body tremors, prostration) and Type II (CS syndrome - coarse tremors, clonic seizures, sinuous writhing, profuse salivation) poisoning following acute toxic doses to rats?
   A) carbamates
   B) organochlonines
   C) cyclodienes
   D) pyrethroids
   E) organophosphates

2. Which of the following is NOT true regarding the exposure of humans to the industrial solvent tricholoethylene (TCE)?
   A) via inhalation, it is readily and nearly completely absorbed into the pulmonary system
   B) via ingestion, it crosses the gastrointestinal mucosa by passive diffusion and has very high bioavailability
   C) dermal exposure results in extensive absorption even with brief exposure
   D) it is quickly distributed within the body once absorbed
   E) all of the above

3. Which of the following is NOT true regarding methyl tertiary butyl ether?
   A) used as a gasoline additive
   B) identified as a cause of leukemia in humans
   C) imparts an unpleasant taste and odor to contaminated drinking water
   D) produces renal and testicular tumors in rats
   E) an example of a compound for which the relevance to humans of animal cancer bioassay results have been questioned

4. Which of the following is NOT produced by fungi?
   A) aflatoxin
   B) amanitin
   C) phalloidin
   D) saxitoxin
   E) ergot

5. Which of the following is NOT a mycotoxin?
   A) ergot alkaloids
   B) tetradotoxin
   C) ochratoxin
   D) penicillinic acid
   E) zearalenones

6. Which of the following is true regarding ricin?
   A) is derived from the plant *Rhamnus purshiana*
   B) is a single, amino acid chain lectin with a molecular weight of ~65,000
   C) the toxin is largely destroyed in the human gastrointestinal tract
   D) effects of its ingestion are primarily related to central nervous system stimulation
   E) its mechanism of toxicity is similar to that of esculin found in horse chestnut

7. Which of the following is likely responsible for nausea, vomiting, diarrhea, tingling and numbness of the lips and tongue, muscle aches and metallic taste that occurs about 6 hours following a banquet?
   A) *Staphylococcus aureus*
   B) organophosphate insecticide residues on foods
   C) ciguatoxin
   D) *Salmonella* sp.
   E) *Clostridium perfringens*

8. Which metal is most likely to present a hazard in an electron microscopy laboratory?
   A) gold
   B) osmium
   C) iridium
   D) platinum
   E) cadmium

9. Which of the following is NOT true for second generation rodenticides, such as brodifacoum?
   A) were developed because some strains of rats have become genetically resistant to warfarin
   B) stimulate prothrombin synthesis
   C) act by inhibiting Vitamin K dependent factors (VII, IX and X)
   D) have a rapid onset of acute signs of toxicity
   E) have a long half-life and require lengthy therapy for accidental human exposure

10. Which of the following is the most likely explanation for renal failure with calcium oxalate crystals present in the proximal tubules of a domestic cat?
   A) warfarin-based rat poison
   B) propylene glycol-based antifreeze
   C) sodium hypochlorite solution
   D) ethylene glycol-based antifreeze
   E) leaves from a dumb cane plant

11. Which of the following is most acutely toxic to humans?
   A) pyrethroid ester
   B) chlorophenoxy acids
   C) soman
   D) DDT
   E) Ivermectin

12. Inorganic arsenic produces all of the following toxic effects EXCEPT
A) nervous system abnormalities
B) skin lesions
C) hemolysis
D) peripheral vascular disease
E) fever

13. Which of the following is NOT involved in reactions involved in photochemical air pollution?
A) carbon monoxide
B) ozone
C) nitrogen dioxide
D) peroxyacetyl nitrate
E) aldehydes

14. Which of the following is/are true regarding organomercurial poisoning?
A) causes psychomotor retardation from prenatal exposure
B) could be produced by consumption of seafood
C) produces neurotoxicity
D) was reported in Minamata Bay
E) is hycholyzed by sediment bacteria

15. Which of the following is true?
A) in the kidney, cisplatin primarily targets the glomerulus
B) asthmatic syndrome has been reported in individuals exposed to platinum dust
C) the antitumor effects of platinum complexes are due to persistent inhibition of RNA synthesis
D) the use of cisplatin as an antitumor agent is limited by cardiotoxicity
E) platinum complexes are not mutagenic

16. All of the following are true regarding dermal application of chemicals EXCEPT
A) absorption is limited primarily by the stratum corneum
B) hair follicles and other gaps in the epidermis have a minor role in absorption
C) molecular weight and partition coefficient of the compound affect the rate of absorption
D) diffusion through lower layers of the stratum granulosum depends on the blood flow and interstitial fluid movement
E) the absorption of highly lipophilic compounds, such as TCDD, through the skin is limited by their relative insolubility in phospholipid bilayer

17. Which of the following can be predicted about the effects of radon exposure in the home based on epidemiological studies of radon exposures in miners?
A) residents in high-rise apartments would be predicted to be exposed to higher levels of radon than those in ground level apartments
B) short exposures to very low levels of radon and its daughters would be predicted to produce a clear excess of lung cancer in homeowners
C) given similar levels of radon exposure, smokers would have a similar overall lifetime lung cancer risk compared to nonsmokers
D) smaller particles can increase the actual dose per unit exposure delivered by radon to bronchial tissue
E) urban areas generally have larger percentages of their residents with high radon exposures relative to suburban areas

18. Which of the following is NOT true regarding the differences between carbamates and organophosphates?
A) unlike inhibition by organophosphates, inhibition of acetylcholinesterase by carbamates is readily reversible
B) carbamates are more rapidly biotransformed in mammalian systems
C) the depression of plasma acetylcholinesterase is not a good indicator of the severity of poisoning by carbamates
D) the mechanism by which carbamates and organophosphates elicit their toxicity is identical, however, organophosphates toxicity tends to be more persistant in duration
E) unlike organophosphate poisoning, there is no benefit from treating carbamate intoxication with atropine

19. The most common route(s) of exposure of agricultural workers to pesticides is/are
A) oral
B) dermal
C) inhalation
D) ocular
E) inhalation and oral

20. Which of the following is the most effective treatment to prevent thyroid cancer from exposure to ^{131}I following a nuclear reactor accident?
A) reduced glutathione (GSH)
B) potassium iodide (KI)
C) methylene blue
D) heavy water (D_2O)
E) thyroid hormone

21. Which of the following statements best describes metal fume fever?
A) direct pulmonary toxicity occurs and serum metal levels are elevated
B) a recurrent flu-like syndrome that develops shortly after exposure to metal oxide fumes
C) exacerbation of symptoms during the workweek is typical
D) the drug of choice in management is dimercaptopropanol (BAL)
E) chest radiography during the episode usually reveals patchy pulmonary infiltrates

22. Which of the following is NOT true regarding vinyl chloride?
A) induces fibrosis
B) is associated with scleroderma-like syndrome
C) is an inducer of cytochrome P450
D) is associated with angiosarcoma of the liver in humans
E) is associated with lung cancer

23. Which of the following is used in radiation therapy machines for treating tumors?
A) iodine 137
B) cobalt 60
C) gold 198
D) chromium 51
E) phosphorus 32

24. Hydrogen cyanide
A) binds to reduced heme
B) is less toxic than thiocyanate
C) is inactivated primarily by Phase I metabolism
D) binds to methemoglobin
E) is converted to SCN- by diaphorase

25. Which of the following is NOTan adverse effect of prolonged pharmacological corticosteroid treatment?
A) cataracts
B) intravascular thrombosis
C) myopathies
D) osteoporosis
E) peptic ulcers

26. Each of the following antibiotics is identified with the correct toxicity EXCEPT
A) chloramphenicol: pancytopenia
B) penicillins: hypersensitivity reactions
C) aminoglycoside antibiotics: nephrotoxicity
D) cephalosporins: ototoxicity
E) tetracyclines: phototoxicity

27. Cardiotoxicity produced by cisapride is due primarily to
A) chelation of calcium ions
B) generation of oxygen-derived free radicals
C) inhibition of calcium transport across the sarcolemma
D) blockade of potassium hERG channels
E) inhibition of mitochondrial respiration

28. Which of the following least affects intestinal absorption?
A) intestinal motility
B) intestinal lymph flow
C) integrity of intestinal mucosa
D) rate of ingestion
E) systemic blood pressure

29. The term "homogeneous and polydispersed aerosol' refers to which of the following?
A) one chemical - geometric standard deviation less than 1.2
B) different chemicals - geometric standard deviation greater than 1.2
C) one chemical - geometric standard deviation greater than 1.2
D) different chemicals - geometric standard deviation less than 1.2
E) one chemical - geometric standard deviation not measurable

30. Which of the following is true regarding allergic contact dermatitis?
A) is primarily due to skin irritation
B) a greater response is produced at challenge than at initial contact
C) large molecules and proteins are most commonly implicated
D) involves histamine release by mast cells
E) a positive skin response on initial contact indicates sensitivity

31. Which of the following is true regarding ototoxicity produced by gentamicin?
A) loss of sensory hair cells is reversible
B) younger patients are most susceptible
C) amount of damage correlates with accumulation of drug in the inner ear
D) tinnitus rarely occurs
E) low frequency range is affected first

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

32. Which of the following has/have been associated with occupational exposures to various organic solvents and solvent mixtures?
1) acquired dyschromatopsia (color vision loss)
2) miosis
3) decreased visual contrast sensitivity
4) toxic amblyopia

33. Which of the following is/are true?
1) serpentine asbestos fibers are more soluble in tissue than are amphibole asbestos fibers
2) epidemiological evidence indicates that amphibole asbestos fibers are more potent than chrysotile asbestos fibers in the induction of fibrotic lung disease and lung cancer
3) the incidence of lung cancer in asbestos workers is higher in smokers than in nonsmokers
4) chrysotile asbestos fibers are cleared from the lungs more rapidly than are amphibole asbestos fibers

34. Which of the following is/are general characteristics of atmospheric particles?
1) diffusion is the predominant mechanisms of deposition of ultrafine particles
2) a concentration of ultrafine particles that would be used for an acute limit toxicity study agglomerates less than aerosols of one to four microns
3) "coarse mode" particles in ambient atmosphere tend to travel shorter distances (<1 to 10 kilometers) than "fine mode" particles (100+ kilometers) due to sedimentation processes
4) mechanical crushing and resuspension of dusts are the primary conditions under which ultrafine particles are formed

35. Which of the following is/are true regarding the human respiratory tract?
1) inhaled particles with diameters of 2 nanometers or less deposit primarily in the nasal passages
2) inhaled particles with diameters of 10 micrometers or greater deposit primarily in the nasal passages
3) for very large particles, greater than 50 micrometers, the probability of being inhaled is low
4) minimal deposition of particles of any size occurs in the tracheobronchial region since the inhaled particles are either removed by the nose or are transported to the alveoli

36. Which of the following is/are correct concerning the management of intoxication from organophosphate or carbamate pesticides?
1) atropine, by competitively antagonizing acetylcholine at muscarinic receptors, is indicated for the treatment of organophosphate or carbamate intoxication
2) atropine and pralidoxime (2-PAM) share the same mechanism and are additive in their antidotal efficacy to counteract the toxicity of organophosphates or carbamates
3) carbamates are poor substrates for acetylcholinesterase and produce symptoms for no more than 24-48 hours, while organophosphates can irreversibly bind to acetylcholinesterase creating signs and symptoms of intoxication that can persist for several weeks
4) the hydrolytic regeneration of phosphorylated acetylcholinesterase to the free, active enzyme by 2-PAM is independent of the residence time of the organophosphate at the active site on the enzyme

37. Chelating agents used for the treatment of mercury poisoning include
1) Dimercaptopropanol (BAL)
2) Penicillamine
3) N-acetylcysteine
4) 6,8-Dithiooctanoic acid

38. Which of the following is/are true about parathion?
1) metabolically activated to a much more toxic derivative
2) less toxic than most other organophosphate insecticides such as malathion, dipterex and chlorpyrifos
3) lipid soluble and thus penetrates into the CNS
4) leads to an "instantly aged" acetylcholinesterase

39. Naturally-occurring radioactive material may be found in which of the following?
1) underground mines
2) used oil field drilling pipes
3) certain welding rods
4) the human body

40. Which of the following is/are true regarding tetrodotoxin?
1) found in the lowest concentrations in blood and muscle in puffer fish
2) the concentration is related to the reproductive cycle of the puffer fish
3) also called "fugu" poison
4) acts on the central and peripheral nervous systems

41. Which of the following is/are true regarding the disorientation, giddiness and confusion following acute exposure to solvents?
1) the rapid acute narcotic effects of many solvents suggests rapid metabolism
2) equimolar concentrations of different solvents results in equivalent intensity of narcotic effects
3) animal models are not sensitive enough to demonstrate solvent-induced neurobehavioral effects
4) acute solvent-related CNS effects are generally reversible

42. Which of the following is/are components of organomercurical poisoning ?
1) ataxia
2) paresthesia
3) constriction of visual field
4) anemia

43. Which of the following is/are true regarading arsenic intoxication?
1) arsenic is an element found in most soils, and therefore trace amounts of arsenic are normally found in tissues of domestic animals
2) inorganic arsenic and some organic arsenicals produce similar clinical signs, primarily associated with gastrointestinal lesions
3) inorganic arsenic may occur as trivalent (commercial) or pentavalent (natural) forms
4) pentavalent arsenic is five to ten times more toxic than the trivalent form

44. Ingredient(s) not considered GRAS by the US Food and Drug Administration which is/are added intentionally to food is/are
1) indirect food additives
2) direct food additives
3) dietary supplements
4) color additives

45. Which of the following is/are true regarding the carcinogens formed from nitrosamines?
1) formed from the reaction of amines with oxides of nitrogen
2) metabolically activated by hydroxylation at an alpha-carbon
3) formation can be inhibited by certain vitamins such as C and E
4) in humans, the primary site for formation is in the liver

46. The use of ephedra alkaloids has been banned in the U.S. due to which of the following?
1) induction of cardiac arrhythmias
2) increased risk of stroke
3) increased blood pressure
4) increased risk of tinnitus

47. The mechanism of perchlorate toxicity includes which of the following?
1) competitive inhibition of the uptake of iodine by the thyroid
2) increases in circulating levels of T4 and T3
3) the development of goiter
4) compensatory decrease in the secretion of TSH by the pituitary gland

48. Which of the following statements is/are true about organophosphate-induced delayed neuropathy (OPIDN)
1) similar duration and intensity of toxicity as for carbamate poisoning
2) associated with exposure to tri-o-tolyl phosphate (TOTP)
3) agents that inhibit neuropathic target esterase (NTE) also inhibit OPIDN
4) involves an initial muscle weakness in the arms and legs

49. Lead poisoning commonly produces symptoms in which of the following?
1) renal system
2) neuromuscular system
3) central nervous system
4) skeletal system

50. Which of the following is/are true?
1) trivalent chromium is an essential trace element and is often found in dietary supplements
2) epidemiological studies have shown that hexavalent chromium produces tumors in the human respiratory tract
3) chromium containing alloys (e.g., bracelets) can produce contact dermatitis
4) in the stomach, hexavalent chromium is rapidly oxidized to trivalent chromium

ABT RECERTIFICATION EXAMINATION #23

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Key for the following questions
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1. In a dietary carcinogenicity study the male rats have a mean food consumption of 140 g/rat/week. If the mean body weight of the rats was 500g and the test article was mixed into the diet at 100 ppm what is the mean dose received in mg/kg/day?
   A) 1
   B) 4
   C) 10
   D) 70
   E) 100

2. An increase in thyroid follicular cell tumors was noted in compound-treated rats in a 2 year carcinogenicity bioassay. Serum from the 30 day study was assayed with the following noted in the high dose animals: moderately increased T_4 amd TSH. (2x-3x and control) moderates decreased T_3 (0.5x control) markedly increased rT_3 (15x control). Which of the following is the most likely mechanism?
   A) increased thyroid hormone clearance by induction of ODP-glucuronyltransferase (UDPGT)
   B) decreased conversion of T_4 to T_3 by inhibition of a 5' monoiodinase
   C) increased thyroid hormone clearance by displacement from thyroid-binding globulin
   D) inhibition of thyroid hormone synthesis by inhibition of thyroid peroxidase
   E) direct-acting genotoxicity

3. Somatic mutations differ from germ cell mutations in that somatic mutations
   A) are lethal
   B) result from chemical exposure
   C) cannot be transmitted to progeny
   D) are point mutations
   E) are dominant

4. Heinz bodies in erythrocytes are
   A) nuclear fragments
   B) denatured hemoglobin
   C) iron granules
   D) giant lysosomes
   E) protozoan parasites

5. Which of the following is NOT a drug: mechanism pair for immune-mediated thrombocytopenia?
   A) penicillin:hapten directed antibody
   B) heparin:immune complex
   C) chloramphenicol:thrombotic thrombocytopenic purpura
   D) quinidine:acquired antibody to drug-induced epitope
   E) mitomycin:hemolytic-uremic syndrome

6. Which of the following characteristics of an agent would be of most concern for the potential to cause human cancer?
   A) induces a_{2u}-globulin nephropathy in the rat kidney
   B) induces peroxisome proliferation in the mouse liver
   C) is genotoxic in rats
   D) causes thyroid neoplasia in the rat
   E) causes liver hyperplasia in the rat

7. Which of the following is NOT characteristic of tumor induction?
   A) the presence of types of tumors not seen in controls
   B) an increase in the incidence of tumor types that occur in the controls
   C) the development of tumors earlier than in controls
   D) an increased multiplicity of tumors
   E) the absence of tumors in control animals

8. The type of placenta in the human is
   A) epitheliochorial
   B) syndesmochorial
   C) hemochorial
   D) hemoendothelial
   E) thicker than that seen in horses or pigs

9. All chemicals which induce resorption in animals also
   A) are teratogenic
   B) produce abortion in humans
   C) induce chromosomal damage
   D) induce mutatgenic damage
   E) none of the above

10. The renal adenomas/carcinomas seen with chronic exposure to unleaded gasoline
   A) are due to a direct genotoxic effect on renal cellular DNA
   B) are due to peroxisomal proliferation in susceptible renal cells
   C) occur in both sexes of mice and rats
   D) are seen only in male rats and are due to a non-genotoxic (or epigenetic) mechanism
   E) result from an enhanced renal degradation of a protein, alpha$_{2u}$-globulin, that is unique to male rats

11. Acute tubular renal necrosis is associated with excessive exposure to which of the following?
   A) cylosporine
   B) interleukin-2
   C) cadmium
   D) lithium
   E) amphotericin B

12. Chloroform-induced nephrotoxicity
A) has the glomerulus as its primary cellular target
B) is characterized by reduced blood urea nitrogen levels
C) has no gender-related differences in severity
D) is due to the direct effect of chloroform itself
E) has species-related differences in sensitivity

13. Acute exposure to which of the following would result in initial anuria followed by polyuria and a concomitant decrease in renal concentrating capacity?
A) tetracycline
B) mercuric chloride
C) gentamicin
D) cadmium chloride
E) propylene glycol

14. Hematopoietic toxicity from exposure to benzene or benzene metabolites results from
A) changes in hemoglobin structure
B) damage to cellular membranes
C) inhibited growth and development of pluripotent bone marrow stem cells
D) immune-mediated mechanisms
E) activation by bone marrow CYP1A metabolism

15. Which of the following modes of deposition is more significant for inhaled fibers than for particles?
A) sedimentation
B) diffusion
C) interception
D) impaction
E) electrostatic charge

16. The volume of gas remaining in the lungs after normal exhalation is referred to as
A) total lung capacity
B) expiratory reserve volume
C) vital capacity
D) functional residual capacity
E) tidal volume

17. Damage to the hepatic cytoskeleton by microcystin
A) requires biotransformation to yield a covalently bound reactive intermediate
B) produces injury within the Kupfer cell resulting in loss of endothelial wall integrity
C) decreases the rate of phosphorylation of those enzymes that are critical to energy production within the cytoskeletal apparatus
D) occurs in both hepatocytes and Kupfer cells
E) leads to hyperphosphorylation of cytoskeletal elements leading to collapse of the microtubular actin scaffold

18. Which one of the following statements is NOT true?
A) serum ALT activity may moderately increase as a result of enzyme induction
B) serum ALT activity decreases with alterations in hepatocyte permeability
C) the magnitude of ALT increase does not differentiate between reversible or irreversible injury
D) the magnitude of ALT increase roughly parallels the number of cells or amount of hepatic mass affected
E) anticonvulsants, thiacetarsemide, and glucocorticoids are common inducers

19. What is the dominant driving force for systemic uptake of gases?
A) perfusion
B) ventilation
C) acid-base balance
D) diffusion
E) water solubility

20. Phenyl benzyl carbamate inhibits neuropathic target esterase (NTE) *in vitro* and *in vivo*, leading to which of the following best conclusion?
A) it will produce delayed neuropathy
B) it would produce delayed neuropathy if it also undergoes the "aging" reaction
C) atropine must never be given in intoxications with this compound
D) 2-PAM (pralidoxime) must never be given in conjunction with this compound
E) this compound would not be expected to inhibit butyrylcholinesterase

21. Which of the following is NOT correct?
A) exencephaly = defective development of the cranium, allowing the brain to lie outside of the skull
B) internal hydrocephalus = an abnormal accumulation of fluid within and dilation of the ventricles of the brain
C) lissencephalia = decreased depth or absence of cerebral convolutions
D) atresia ani = imperforate anus
E) talipes equinus = kidneys fused medially in the shape of a horseshoe

22. Hemolytic anemia is usually characterized by which of the following?
A) increased numbers of reticulocytes in the circulating blood
B) a reduction in polychromatic erythrocytes in the circulating blood
C) erythroid hypoplasia in the bone marrow
D) a reduction in mean red blood cell volume
E) increased haptoglobin concentrations

23. Which of the following is a characteristic of dendritic cells?
- A) originate from marrow progenitors and subsequently migrate to sites of exposure to foreign antigens
- B) are found in the brain, testis and eye
- C) have minimal interaction with lymphocytes
- D) are derived from neural tissue
- E) are not found in the epithelial surface of the skin, gastrointestinal tract or lung

24. Which of the following is NOT true regarding photoallergy reactions?
- A) are true type I hypersensitivity reactions
- B) are distinct from phototoxicity reactions
- C) are similar to contact sensitization reactions but require UV light
- D) occur only after topical exposure
- E) require prior sensitization

25. Which of the following statements is correct for reproductive responses?
- A) male mammalian reproductive processes are more complex than those in the female
- B) the endocrine system of the female is less complex and dynamic that that of the male
- C) Leydig cells are more sensitive to environmental toxicants than germ cells
- D) unlike the mature oocyte, the mature sperm maintains a DNA repair ability
- E) the gonad itself is not required for altering FSH or LH synthesis at the onset of puberty

26. Which of the following is NOT associated with isoniazid?
- A) treatment of tuberculosis
- B) hydrazine metabolite
- C) peripheral neuropathy
- D) dose-dependent hepatotoxicity
- E) acetylated metabolites

27. Which statement is most correct regarding the liver?
- A) dose dependent cellular injury results only from substances which are directly hepatotoxic
- B) massive injury to the endoplasmic reticulum, specifically in the centrilobular hepatocytes, causes rapid cell death due to necrosis
- C) liver toxicants which are indirect acting can lead to increasing cellular enzyme release by creating localized deficiency states
- D) intramitochondrial calcium release occurs only when the hepatic cell membrane has ruptured
- E) fatty liver as a manifestation of hepatic injury always leads to Kupffer cell necrosis causing serum transaminase elevation

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Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

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28. Excitatory amino acids
- 1) act as neurotransmitters in the central nervous system
- 2) at high doses, can cause neuronal swelling and cell death but seem to spare axons
- 3) induce toxicity similar to that associated with the "Chinese restaurant syndrome"
- 4) are exemplified by the seaweed isolate, kainate

29. Process(es) by which particulate material is removed from the pulmonary region once it has been deposited include(s)
- 1) phagocytosis and clearance via mucociliary action
- 2) phagocytosis and removal via lymphatic drainage
- 3) dissolution of the particles and removal via the bloodstream or lymphatics
- 4) sequestration in pulmonary macrophages

30. Characteristics of the normal immune system include
- 1) memory cells which become activated and generate a large number of responding cells after reexposure to a specific antigen
- 2) B and T lymphocytes which both originate from the same pluripotent stem cells
- 3) natural killer cells which are potentially cytotoxic for certain malignant cells in the absence of prior sensitization
- 4) separate and independent functions of B and T lymphocytes

31. Which of the following accurately represent(s) the Local Lymph Node Assay procedure?
- 1) the method includes both an *in vivo* phase and an *in vitro* phase in which the lymph node cells are cultured
- 2) conduct of the assay requires the use of tritiated thymidine, and a stimulation index of 3-fold (treated group/control group) is accepted as a positive proliferative response
- 3) it has an induction phase as well as a challenge phase in which a dermal hypersensitivity reaction is elicited
- 4) several concentrations of the test substance may be used in order to establish a dose response curve for induction

32. Ultraviolet radiation in the middle range (300nm, UV-B) causes
1) skin cancer
2) sunburn
3) formation of Vitamin D
4) irradiation of subcutaneous blood vessels

33. Zinc fume fever is
1) not associated with elevated body temperature
2) caused by a very fine dust of metallic zinc
3) a chronic disease characterized primarily by pulmonary fibrosis
4) caused by zinc oxide

34. Toluene diisocyanate (TDI)
1) may cause respiratory hypersensitivity
2) may cause contact dermatitis
3) is used in the manufacture of plastics and is highly reactive with amino groups of proteins
4) is used as a fumigant in the control of insects and rodents

35. Acute delayed neurotoxicity studies are required by the U.S. Environmental Protection Agency
1) for all pesticides that are structurally related to organophosphates
2) to include a formalized testing component for behavioral changes
3) to be conducted in hens
4) to include an electrophysiological component of testing

36. Nicotine
1) places an increased burden on the heart through its acceleration of heart rate and blood pressure
2) can cause nausea, vomiting, and headache at toxic doses and respiratory paralysis at lethal doses
3) exerts its effects by binding to a subset of cholinergic receptors located in ganglia of the neuromuscular junction and in the CNS
4) may be associated with a sensation of relaxation and EEG alterations

37. According to U.S. Environmental Protection Agency guidelines, if developmental toxicity and maternal toxicity occur at the same doses of an agent in a developmental toxicology study
1) the agent should not be considered a developmental toxicant
2) the developmental toxicity can be considered secondary to the maternal toxicity
3) the agent may be considered a developmental toxicant if the fetal effects are more severe than the maternal effects
4) the agent can be considered both a developmental and a maternal toxicant

38. Which of the following is/are correct concerning cocaine use in pregnant women?
1) uterine blood flow can be diminished
2) fetal hypoxia can occur
3) the likelihood of placental hemorrhages (abruptions) and miscarriage is increased
4) maternal blood pressure is decreased

39. High doses of aspirin and non-steroidal anti-inflammatory drugs (such as indomethacin) in rats impair delivery, prolong gestation and increase the incidence of neonatal death. These effects are considered to be due to
1) an extension of the pharmacologic effects of the drugs
2) teratogenic effects on the fetus
3) decreased uterine motility
4) decreased ability of blood to clot

40. Administration of a chemical to a pregnant subject during early gestation (preimplantation blastocyst stage)
1) may be embryolethal and kill the blastocyst outright
2) can result in no or slight effects on growth
3) may cause embryotoxic effects such as cell death, but normal offspring may result because of regulative growth potential of cells at this developmental stage
4) does not affect the inner cell mass because of lack of a placenta and its vasculature that delivers the chemical to the conceptus

41. Which of the following statements about Langerhans-dendritic cells is/are correct?
1) can migrate into the lymphatic system where they serve as an antigen-presenting cell in the lymph nodes
2) are bone marrow-derived cell with a similar lineage as the macrophage
3) are found primarily in the epidermis, mucosal epithelium and lymphoid tissues
4) play an important role in immediate hypersensitivity reactions

42. Which of the following is/are true concerning ocular toxicity?
1) systemic administration of 2,4-dinitrophenol may cause cataracts in some individuals
2) methanol intoxication may lead to blindness due to its effect on ganglion cell neurons
3) tamoxifen has resulted in the development of perimacular deposits
4) steroid administration may lead to development of a retinopathy

43. Which of the following measures will minimize potential confounding effects of rodent light-induced retinopathy on the outcome of toxicology studies?
1) ensuring that room light levels do not exceed 325 lux (30 foot-candles) at 1.0 meter above the floor
2) ensuring that a consistent dark cycle (at least 10 hours daily) occurs in rodent rooms
3) rotating or staggering the positions of dose groups within a cage rack
4) supplementation of commercial rodent chow with vitamin E

44. The Endocrine Disrupter Screening and Testing Advisory Committee (EDSTAC) advised the U.S. Environmental Protection Agency that pesticides and other high priority chemicals be screened for disruption of the actions of which of the following hormones?
1) estrogens
2) androgens
3) thyroid hormones
4) corticosteroids

45. The Ames test
1) detects the frequency of reversions or back mutations in auxotrophic bacteria
2) employs several different strains of *Salmonella typhimurium*
3) measures the frequency of histidine-independent bacteria
4) detects mutations in the *hprt* gene

46. Which of the following cell types(s) do/does NOT possess surface receptors for complement?
1) macrophages
2) B-cells
3) NK cells
4) T-cells

47. Which of the following is are true regarding the mechanism of induction of pancreatic nodules, adenomas and carcinoma in rats chronically fed soybean trypsin inhibitor?
1) involves a sustained and enhanced release of cholecystokinin (CCK)
2) involves decreased CCK secretion as a result of the inhibition of trypsin in the gastrointestinal tract
3) is not considered to be relevant to humans
4) does not involve CCK

48. Dose-response relationships in teratology studies are characterized in part by which of the following statements?
1) the response of the embryo can be highly variable and dependent on a number of factors other than dose
2) typically, the dose-response curve for embryotoxic effects has a steep slope, going from minimal to maximal within a discrete range of doses
3) the issue that a safe threshold exists for exposure to teratogenic agents remains controversial and must await studies which separate the low-dose effects of teratogenic agents from the background of "spontaneous" defects in laboratory animal and human populations
4) with increasing dosages, there is no reason to believe that the number of malformed offspring decreases as embryolethality increases

49. Which of the following is/are correct regarding mutagenesis?
1) fluorescence *in situ* hybridization (FISH) is an assay being utilized to detect chromosomal anomalies
2) bacterial cell assays (e.g., Ames test) and mammalian cell assays (e.g. mouse lymphoma tk locus test) are both capable of detecting forward mutations
3) a standard genotoxicity test battery to support the overall registration of pharmaceuticals will customarily include an *in vitro* gene mutation test in bacteria, an *in vitro* cytogenetic evaluation of chromosomal damage with mammalian cells , and an *in vivo* test for chromosomal damage using rodent hematopoietic cells
4) aneuploidy is considered to be a clastogenic change

Key for the following questions
Choose the single most appropriate answer

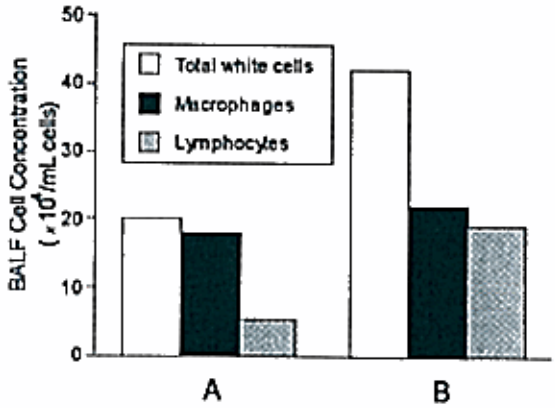

Table 20
The figure depicts bronchiole alveolar lavage fluid (BALF) cell concentration in two populations of workers (A, unexposed and B, exposed) in a high technology ceramics plant. The bars represent the folloing cell types recovered from BALF.

50. Refer to Table 20
What compound is implicated in the etiology for such an observation?
A) gallium arsenide
B) beryllium
C) asbestos
D) tungsten
E) technetium

ABT RECERTIFICATION EXAMINATION #23

May 5, 2006 *PART C*

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1997 and 2002. There are three parts to the exam. An answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading by **June 30, 2006**. If you wish to return them by courier, the physical address is 9200 Sanctuary Court, Raleigh, NC 27617. If you do not achieve a passing score on any of the parts, you will be contacted and given another opportunity to pass. If you pass all three parts, your scores will be provided to you in November of this year, along with the results of the Recertification Committee's review of your other application materials. If you have any questions please contact the ABT office at (919) 841-5022 or info@abtox.org.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.

2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.

3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

Key for the following questions
Choose the single most appropriate answer

1. The blood-brain barrier
   A) facilitates the movement of water soluble compounds by paracellular transport
   B) contains multidrug resistant proteins that transport some chemicals into the blood
   C) is consistently effective across all areas of the brain
   D) is fully developed at birth
   E) is comprised of a fenestrated capillary endothelium

2. All of the following are important factors when considering species-to-species extrapolation of toxicity EXCEPT
   A) relative sensitivity of the animal model compared to humans
   B) relative population differences
   C) differences in body weight and body surface areas
   D) relative food consumption of the animal model compared to humans
   E) relative differences in biotransformation pathways

3. Interspecies comparisons for predicting adverse effects in humans are usually best made based on dose relative to
   A) body surface area
   B) body weight
   C) systemic exposure
   D) differences in rates of absorption
   E) volume of distribution

4. Which of the following statements regarding CYP3A enzymes is correct?
   A) induced by phenytoin
   B) inhibited by oemeprazole
   C) CYPA3 activity demonstrates little inter-individual variability
   D) inducers of the enzymes in one species of animals will induce the enzymes in all other species
   E) are neither present nor inducible in the brain

5. Which of the following is NOT a function associated with the special transport system, p-glycoprotein?
   A) active transport
   B) contributes to the blood-brain barrier
   C) resistance of tumor cells to chemotherapeutic drugs
   D) facilitated diffusion
   E) may export xenobiotics from cells

6. Which of the following is true?
   A) blood-brain barrier consists of endothelium, astrocytes and other components
   B) blood-testis barrier consists of Leydig cells
   C) placental barrier consists of maternal tissues only
   D) placental barrier can eliminate most toxic chemicals by biotransformation, but blood-brain and blood testis barriers cannot
   E) blood-brain barrier is fully developed in newborns

7. Which of the following regarding toxicokinetics is NOT true?
   A) it is the study of the modeling and mathematical description of the time course of disposition of xenobiotics in the whole organism
   B) the apparent volume of distribution (Vd) is the apparent space into which an amount of chemical is distributed in the body to result in a given plasma concentration
   C) the half-life of elimination (T 1/2) is the time required for the blood or plasma chemical concentration to decrease by one-half
   D) bioavailability (F) following an oral exposure can be determined as: the area under the oral plasma concentration versus time curve (AUC_{po}) divided by the oral dose ($Dose_{po}$)
   E) clearance describes the rate of chemical elimination from the body in terms of the volume of fluid containing chemical that is cleared per unit of time

8. Which of the following is true?
   A) pretreatment with 2-butanol protects rats from carbon tetrachloride-induced hepatotoxicity
   B) ethanol intake near the time of exposure to methanol increases the likelihood of methanol poisoning
   C) co-exposure to benzene and toluene increases the genotoxicity and hematological toxicity of benzene
   D) workers in an isopropanol bottling plant developed hepatotoxicity upon exposure to a normally non-hepatotoxic concentration because isopropanol is metabolized to acetone, an inducer of CYP2E1
   E) repetitive alcohol exposures may induce CYP2E1 activity and increase the clearance of other solvents

9. Which of the following reactions is microsomal?
   A) aldehyde oxidation
   B) sulfate conjugation
   C) monoamine oxidation
   D) glucuronide conjugation
   E) sulfoxide reduction

10. Which compound is contraindicated in the treatment of *Amanita muscara* poisoning?
A) atropine
B) cimetidine
C) cyclosporins
D) penicillin
E) gentamicin

11. Which of the following is NOT true about occupational exposure limits?
A) are often based on limited experimental data or chemical studies
B) epidemiological data are required
C) are more valid if based on dose-response studies
D) can be validated based on the results of workplace surveillance programs
E) can be established as enforceable standards

12. Major manifestations of fluoride toxicity are all of the following EXCEPT
A) osteosclerosis
B) mottled dental enamel
C) dental fluorosis following postnatal fluoride ingestion
D) elevated osteoblast activity
E) replacement of hydroxyapatite with fluoroapatite

13. Damaged proteins are NOT repaired or targeted for removal by
A) the thioredoxin pathway
B) the glutaredoxin pathway
C) stress or "heat shock" proteins
D) recombination repair
E) ubiquitination

14. The U.S. Occupational Safety and Health Hazard Communication Standard
A) limits the airborne concentration of a material to which a person may be exposed to during a 12-hour period
B) regulates the way in which the chemical is used in processing
C) is not specific for signs posted in work areas where the chemical is used
D) specifies that information for all toxic chemicals present at >1% be included in MSDS
E) specifies that carcinogens identified by IARC be included on the product label if present at 1% or greater

15. Which of the following does NOT facilitate distribution of xenobiotics to a target organ?
A) permeability of the capillary endothelium
B) specialized transport across the plasma membrane
C) binding to plasma proteins
D) reversible intracellular binding
E) accumulation in cell organelles

16. Which of the following pairings of human polymorphic metabolic activity and compound classes is NOT correct?
A) acetylation (N- acetyl transferase, NAT12) - heterocyclic amines (bioactivation of aromatic amines)
B) methylation (catechol-O-methyltransferase, COMT) - estrogen, L-Dopa
C) methylation - arsenic/selenium
D) conjugation (glutathione-S-transferase, GST) - aromatic and alkene epoxides
E) oxidation (cytochrome P450 enzymes) - many different drug classes

17. Phosphine (PH_3) is
A) a hazard in coal mines unless there is adequate ventilation
B) similar to phosgene in being a renal toxicant
C) a gas used to fumigate grains, which causes respiratory symptoms
D) a hazard in phosphate mining operations
E) produced when aluminum phosphide reacts with carbon dioxide

18. Which is NOT true regarding the unpaired Student's t-test?
A) the Student's t-test should only be used to test normally distributed populations
B) pairs of groups of only continuous and randomly distributed data should be compared with this test
C) the Student's t-test is particularly useful for testing paired observations and ranked data
D) when more than two groups of data are to be compared, a more stringent adjustment to the probability level (p) is required
E) the two valid variations of the student's t-test are: the modified t-test when group sample sizes are equal but variances are NOT homogeneous, and the Cochran t-test for unequal sample sizes and heterogeneous variances

19. The Standardized Mortality Ratio (SMR) is the ratio of
A) mortality from a specific cause in one standard population to the mortality in another standard population under study
B) mortality rate for the disease under study to the mortality for another standard disease
C) the number of deaths observed in the study population to the number of deaths expected in the study population if it had the same rate structure as the standard population
D) the expected number of deaths in the study population at two different points in time
E) the expected number of deaths in the study population to the actual number of deaths in the standard population

20. In which case has food NOT been shown to be a vehicle for a toxicant in an episode of poisonings?
- A) cadmium in rice causing *itai-itai* disease involved bone pain from osteomalacia
- B) organotin compounds in beer caused renal failure
- C) polychlorinated biphenyls (PCBs) in cooking oils produced chloracne
- D) zearalenone in grains produced a hyper-estrogenic state and decreased reproduction in swine
- E) aflatoxin in grain and seed crops caused liver failure

21. Which of the following statements is NOT correct regarding the drug approval process in the U.S.?
- A) an Investigational New Drug (IND) application seeks approval to start human testing of a new drug or biologic
- B) Phase 1 clinical trials typically are designed to evaluate the safety of a new drug or biologic in humans
- C) Phase 4 clinical trials may be required post-approval to confirm the efficacy and safety, provide dose-response data or provide additional drug interaction data for a new drug or biologic
- D) carcinogenicity studies and developmental toxicity testing are routinely required prior to initiating Phase 1 clinical trials
- E) depending upon the disease being studied, either healthy volunteers or patients with the disease of interest may be enrolled in Phase 1 studies

22. Drug A has an LD_{50} of 600 mg/kg and a no observed effect level of 100 mg/kg in a rat 28 day study. In clinical trials the ED_{50} was found to be 50 mg/kg and adverse effects were seen at 100 mg/kg. The therapeutic index for this drug is
- A) 2
- B) 1
- C) 0.5
- D) 12
- E) 6

23. Under the U.S. Toxic Substances Control Act, the Premanufacturing Notice submitted to the Environmental Protection Agency is required to include all of the following EXCEPT
- A) changes in proportions of chemical mixtures used
- B) molecular structure of chemicals produced
- C) description of byproducts resulting from use
- D) occupational exposures including estimated number of people to be exposed in the future
- E) estimate of the amount of the chemical to be produced in the future

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Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

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24. Indicators of immunosuppression observed in standard nonclinical toxicology studies can include
- 1) hematological changes
- 2) decreased serum immunoglobulin
- 3) alterations in histology of the spleen, lymph nodes, and bone marrow
- 4) increased incidence of tumors

25. Characteristics of toxin-induced apoptosis include
- 1) nuclear changes such as chromatin condensation and fragmentation
- 2) internucleosomal DNA cleavage
- 3) morphological changes including membrane blebbing
- 4) cell swelling leading to leakage of its constituents into the surrounding extracellular space

26. Which of the following is/are true about genetic polymorphisms involved in the response to drugs and chemicals?
- 1) primaquine sensitivity, caused by glucose-6-phospate dehydrogenase (G6PD) deficiency among African and Mediterranean people, is one of the first and best known examples of genetically-based drug sensitivity
- 2) a slow metabolic variant of alcohol deydrogenase is responsible for the facial flushing seen as part of the alcohol intolerance found in some Asian populations
- 3) cytochrome P450 polymorphisms may either enhance or reduce drug efficacy depending on the drug and the CYP form
- 4) fast acetylators are at increased risk for bladder cancer from arylamine exposure

27. Which of the following is/are true?
- 1) apparent volume of distribution changes linearly with dose by first order kinetics
- 2) if biotransformation pathways are saturated, clearance decreases with increasing dose
- 3) compounds which have zero-order kinetics demonstrate dose dependent half-lives
- 4) half-life and clearance are not dose dependent for compounds which follow first order kinetics

28. Which of the following require(s) NADPH-dependent metabolism to cause lung injury?
1) 4-ipomeanol
2) naphthalene
3) paraquat
4) carbon tetrachloride

29. Which of the following may be used prophylactically to protect humans against lethal exposure to potent organophosphate toxicants such as VX or soman?
1) physostigmine
2) atropine
3) human butyrylcholinesterase
4) diazepam

30. Which of the following is/are true of mathematical models used to estimate human risk?
1) the LMS model is appropriate for estimating low-dose carcinogenic potency
2) the Weibull model assumes once exposure occur, risk continues to accrue despite the cessation of exposure
3) the multi stage model incorporates cell-cycle dependent processes and birth/death rates of cells
4) the Moolgavkar, Venson, and Knudson (MVK) model is a biologically-based model which reproduces the multi-stage characteristics of neoplastic development

31. Which of the following is/are correct regarding Structure Activity Relationships (SAR) or Quantitative Structure Activity Relationships (QSAR)?
1) information from QSAR work is advocated by animal welfare groups to minimize the use of animals in testing programs
2) the premise of any QSAR is to relate various physiochemical properties to a biological response in a predictable manner
3) skin and eye irritation are two of the toxicological responses for which SAR or QSAR models have been undergoing development in recent years
4) utility of a QSAR model may be limited when applied to structurally diverse data sets

32. Cyanide
1) toxicity occurs as a result of exposure to inorganic, but not organic, nitriles
2) has a very high affinity for iron in the ferric state
3) reacts readily with microsomal cytochrome oxidase
4) is enzymatically detoxified by rhodanase

33. In performing human health risk assessments, the reference dose (RfD)
1) is used to evaluate cancer risk for some carcinogens
2) is used to evaluate noncancer endpoints
3) assumes there is no dose below which adverse effects are unlikely
4) can be derived from a no adverse effect level by dividing by an uncertainty factor

34. The concept of hormesis
1) can be demonstrated by extrapolation from responses at high doses
2) is demonstrated by a U- or inverted U-shaped dose-response curve
3) is demonstrated by a threshold dose-response curve
4) has been suggested as applicable for carcinogens and noncarcinogens

35. Which of the following is/are requirements of the 1996 US Food Quality Protection Act amendment of the US Federal Insecticide, Fungicide and Rodenticide Act and the US Federal Food, Drug, and Cosmetic Act?
1) assessment of available information about food consumption patterns among infants and children that are likely to result in disproportionately high consumption of foods containing or bearing pesticide residues
2) assessment of available information concerning the special susceptibility of infants and children to the pesticide residues, including neurological differences between infants and children and adults, and effects of *in utero* exposure to pesticide chemicals
3) assessment of available information concerning the cumulative effects on adults, infants and children of pesticide residues and other substances that have a common mechanism of toxicity
4) must ensure that there is a reasonable certainty that no harm will result to infants and children from aggregate exposure to a pesticide chemical residue

36. The 1958 Amendment to the US Food, Drug and Cosmetic Act is significant to regulatory toxicology because the Amendment
1) includes the "Delaney clause"
2) requires that any additive to food must be found "reasonably certain to be safe" before the US Food and Drug Administration may approve its use
3) stipulates that no food additive can be found "reasonably certain to be safe" if the substance has been found to cause cancer in humans or in experimental animals
4) exempts from regulatory oversight substances that are "generally recognized as safe"

37. A child presents to the emergency room following consumption of a bottle of aspirin tablets. Which of the following is/are expected serum chemistry findings?
1) high anion gap
2) decreased serum bicarbonate
3) increased serum potassium
4) increased arterial pCO_2

38. Descriptive epidemiological studies differ from analytic epidemiological studies in that they
1) include case-control studies
2) include cohort studies
3) include intervention studies
4) are conducted to formulate hypotheses

39. Which of the following represent(s) an appropriate pairing of a genetically-modified mouse model and a compound type?
1) LacZ transgenic mouse (MutaMouse) and alkylating agents
2) XPA-deficient mouse and DNA crosslinking agents
3) p53+/- mouse and either genotoxic or nongenotoxic agents
4) TG.Ac mouse and liver tumor promotors

40. Chronic exposure to acrylonitrile
1) results in an increased frequency of renal calculi
2) does not result in an excess cancer rate in exposed workers
3) causes retrolental fibroplasia
4) produced forestomach tumors in rats exposed via drinking water

41. Which of the following is/are true?
1) beryllium metal and all forms of beryllium, including beryl ore, have been associated with disease in humans
2) beryllium disease may develop 20 to 25 years after the last known exposure
3) urinalysis to assess past exposure is not worthwhile because beryllium is detected in the urine only for a few days after exposure
4) families of beryllium workers were historically at increased risk of developing beryllium disease

42. Which of the following is/are among the most common toxicants encountered in the emergency room?
1) barbiturates
2) ethanol
3) acetaminophen
4) pseudoephedrine

43. Occupational asthma is characterized by
1) the presence of variable airflow limitation and reduced airway responsiveness
2) appearance only after a latency period
3) activation of pre-existent asthma by physical stimuli in the workplace
4) possible coexistence of immunologic and non-immunologic components

44. Which of the following can be determined for a sample size of 10 which has a mean of 55 and a standard deviation of 7?
1) the standard error of the mean is 2.2
2) the data are normally distributed
3) the coefficient of variance is 13%
4) the median is 62

45. Measures used to enhance elimination in poisoning include
1) diuresis or increased urine flow
2) the use of pH adjustment for "ion trapping"
3) dialysis either peritoneal or hemodialysis
4) charcoal instead of emesis

46. Which of the following is/are performed by cytochrome P450 enzymes?
1) conversion of arachidonic acid to eicosanoids
2) oxidation of hemoglobin
3) aromatization of testosterone
4) de-iodination of thyroxin

47. The primary target organ(s) of toxicity observed clinically with statins (HMG-CoA reductase inhibitors) are
1) muscle
2) pancreas
3) liver
4) brain

48. Which of the following is/are true?
1) the burning of fossil fuels and the smelting of metals release sulfur dioxide and various particulates
2) ozone is the primary oxidant of concern in photochemical smog
3) sulfur dioxide is predominantly an upper airway irritant which can stimulate bronchoconstriction and mucus secretion
4) because of its high water solubility, ozone penetrates deeply into the lung

49. Which of the following is/are correct regarding occupational asthma?
1) exposure to alpha-amylase in bakeries and proteases in detergent production factories poses a risk
2) laboratory technicians are at risk from exposure to animal urine and dander
3) variable air flow limitation and/or airway hyper-responsiveness are characteristics
4) the presence of arthropods is not associated with work-related asthma

50. Emetics are contraindicated in which of the following situations?
1) if the patient is comatose or only semiconscious
2) if the patient has taken strychnine
3) after ingestion of strong acid
4) after ingestion of petroleum distillates